

Genetic Markers of Critical Illness: Explainable Genomes using Statistical Analysis and LLMs

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Abstract

Genome-wide association studies (GWAS) have significantly advanced our understanding of complex diseases by linking common genetic variants to clinical outcomes. This study investigates the GenOMICC international metadata release, a large-scale, open-access GWAS dataset focused on critically ill COVID-19 patients, with the goal of identifying genome-wide significant variants associated with severe disease progression. We further develop a basic classification model to predict disease severity based on individual genomic profiles. To enhance interpretability and user engagement, we integrate the output of this classifier with Deepseek R-1, a large language model (LLM), prompting it to initiate a natural-language dialogue with the user. This system highlights clinically relevant alleles, explains their potential impact on COVID-19 severity, and enables interactive genomic education while appropriately recommending that detailed diagnostic concerns be discussed with a licensed genetic counselor. Our approach demonstrates the potential of combining statistical genomics with explainable AI to support both clinical risk stratification and patient-facing genetic literacy tools.

CCS Concepts

• **Applied computing** → **Life and medical sciences**; *Genomics*; *Bioinformatics*.

Keywords

GWAS, COVID-19, genetic markers, critical illness, explainable AI, GenOMICC, gene, allele, LLM, AI

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1 Introduction

Genome-wide association studies (GWAS) have emerged as powerful tools in modern genomic medicine, enabling researchers to identify genetic variants associated with complex disease phenotypes across diverse populations. The COVID-19 pandemic has underscored the importance of understanding genetic susceptibility to severe illness, as individuals infected with the same pathogen experience dramatically different clinical outcomes ranging from asymptomatic infection to critical respiratory failure and death.

This study leverages the GenOMICC (Genetics Of Mortality In Critical Care) international metadata release, a comprehensive open-access GWAS dataset specifically focused on critically ill COVID-19 patients. By analyzing this data, we aim to identify specific genome-wide significant variants that may predispose individuals to severe disease progression when infected with SARS-CoV-2. Beyond mere identification, we seek to contextualize these variants within biological pathways and clinical implications.

A significant challenge in genomic medicine is translating complex statistical associations into actionable insights for clinicians and accessible information for patients. To address this gap, we develop a classification model that predicts disease severity based on individual genomic profiles. We then enhance this model's utility by integrating it with Deepseek R-1, a large language model (LLM) capable of natural language dialogue. This integration transforms technical genomic information into conversational explanations of clinically relevant alleles and their potential impacts.

The potential applications of this work extend beyond COVID-19 to the broader field of critical care medicine. By identifying genetic markers associated with severe outcomes, clinicians may develop improved risk stratification protocols and personalized therapeutic approaches. Additionally, our LLM-enhanced system demonstrates how advanced AI can improve genetic literacy while maintaining appropriate boundaries by recommending consultation with genetic counselors for specific diagnostic concerns.

This research addresses the pressing need for explainable genomic medicine in the era of precision health. By combining rigorous statistical methods with innovative AI communication tools, we aim to advance both clinical understanding of genetic risk factors and patients' engagement with their genomic information. The insights gained may contribute to more targeted interventions and improved outcomes for critically ill patients, while establishing a framework for responsible AI-assisted genomic interpretation that could be applied to numerous complex diseases.

2 Related Work

Our work builds upon three key research areas: GWAS studies identifying COVID-19 severity markers, clinical applications of genetic risk prediction, and AI integration for genomic interpretation.

Païro-Castineira et al. [Païro-Castineira et al. 2020] established the foundation through the GenOMICC (Genetics Of Mortality In Critical Care) study, identifying key variants in IFNAR2 and OAS genes associated with severe COVID-19 outcomes. Their subsequent work [Païro-Castineira et al. 2023] expanded this analysis through meta-analysis, identifying 49 genetic variants across diverse ancestral backgrounds, providing crucial insights into biological mechanisms of severe disease including interferon signaling pathways and lung epithelial cell function.

Translating genetic findings into clinical applications, Dite et al. [Dite et al. 2021a,b] developed and validated integrated clinical and genetic models for predicting severe COVID-19 risk. Their approach combined traditional clinical risk factors with genetic markers to create population-based risk stratification tools, demonstrating the potential for implementing genomic information in clinical decision-making.

The integration of AI with genomic data represents our third foundation. Lee et al. [Lee et al. 2021] pioneered explainable AI methods for genomic data interpretation, developing techniques to make GWAS results more accessible to clinicians. Recent advances in large language models, particularly DeepSeek-R1 [DeepSeek-AI et al. 2025], have created new opportunities for translating statistical genetic associations into clinically meaningful insights through enhanced reasoning capabilities suited for complex biomedical interpretation tasks.

Our current work bridges these research streams by combining statistical genomics with explainable AI to enhance both clinical understanding and patient engagement with genetic information related to critical illness.

3 Methodology

3.1 Data Acquisition and Processing

This study utilized the GenOMICC (Genetics Of Mortality In Critical Care) international metadata release 3, specifically focusing on summary statistics from meta-analyses of genome-wide association studies (GWAS) in critically ill COVID-19 patients. The dataset (GenOMICC_international_meta_release3_critical.tsv.gz) was processed using Python with the pandas library to extract relevant genetic variants.

```
gwas_df = pd.read_csv(file_path, sep='t', compression='gzip')
significant_variants = gwas_df[gwas_df['P.value'] < 5e-8]
```

To ensure statistical rigor, we applied the standard genome-wide significance threshold ($p\text{-value} < 5 \times 10^{-8}$) for variant filtering, which accounts for multiple testing correction across the genome. This yielded 4864 genome-wide significant variants across various chromosomes, with particularly strong signals on chromosomes 3, 6, 19 and 21.

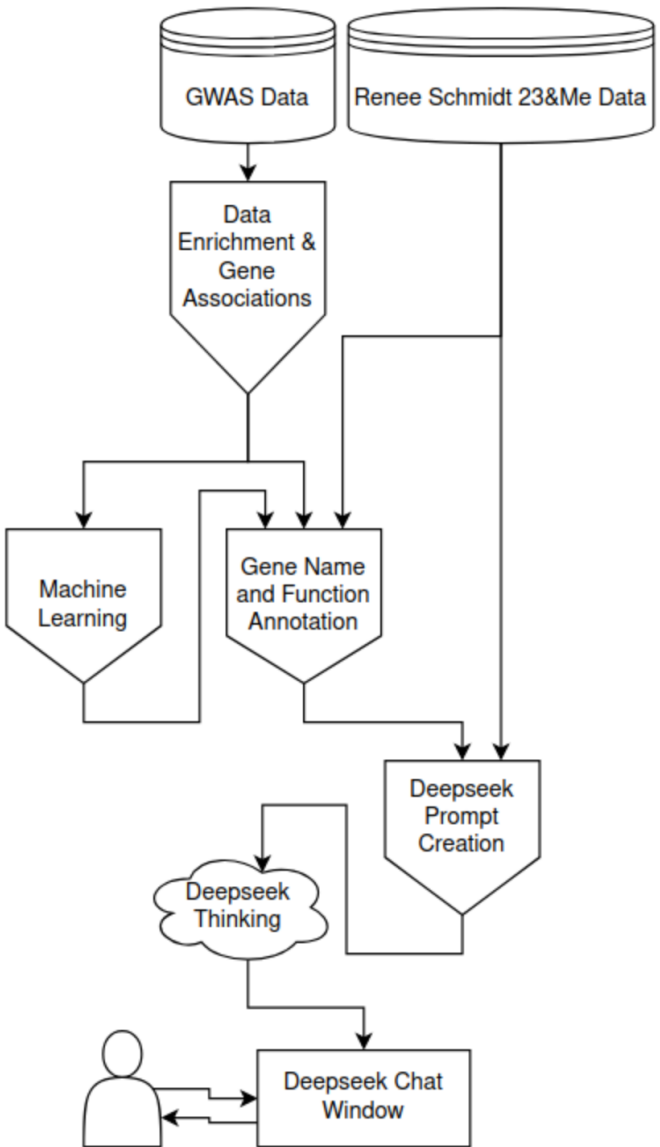


Figure 1: Project Overview Diagram

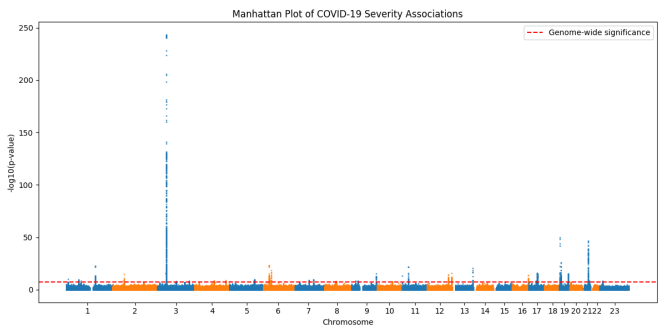


Figure 2: Severity Associations on the Chromosome

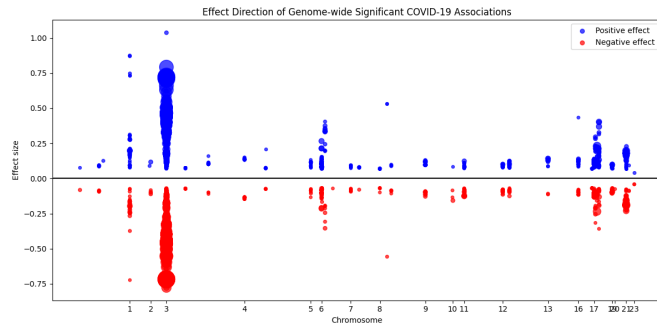


Figure 3: Variant Effects on Covid-19 Outcomes

3.2 Variant Annotation and Functional Mapping

The significant variants were annotated with relevant genomic features using a custom annotation framework. We created a comprehensive dictionary of known COVID-19 associated genes and their approximate genomic locations based on published literature from the GenOMICC studies and related research:

```
covid_gene_associations = {
  ('3', 45800000, 45900000): {
    'gene': 'LZTFL1',
    'function': 'Regulates protein trafficking to ciliary membrane',
    'effect': 'Risk variant increases susceptibility to respiratory failure',
    'mechanism': 'Affects lung epithelial cells and viral entry'
  },
  # Additional gene mappings follow the same structure
}
```

Each variant was mapped to its respective gene using chromosomal location, enabling biological context integration with statistical signals.

3.3 Personal Genome Analysis and Risk Prediction

To develop a personalized risk assessment framework, we implemented a methodology for analyzing individual genomes against the identified significant variants. This involved:

- (1) Loading and processing personal genome data from VCF format
- (2) Matching individual genotypes to significant variants identified in the GWAS
- (3) Calculating personalized effect scores based on genotype and GWAS effect sizes

```
def calculate_personal_effect(row):
    # Parse genotype: 0 = REF, 1 = ALT
    genotype = row['Genotype']

    # Skip complex genotypes or missing data
    if '/' not in genotype and '|' not in genotype:
        return 0

    # Split genotype and count ALT alleles
    if '/' in genotype:
        alleles = genotype.split('/')
    else:
        alleles = genotype.split('|')

    alleles = [int(a) for a in alleles if a != '.']
    alt_count = sum(1 for a in alleles if a == 1)

    # Calculate personal effect: assuming additive model
    return row['Effect'] * alt_count
```

The personal effect calculation implemented an additive genetic model, where the impact of each risk allele accumulates linearly. This allowed us to generate a composite risk score reflecting an individual's genetic predisposition to severe COVID-19 outcomes.

3.4 Comprehensive Risk Profiling and Visualization

We developed a multi-dimensional visualization approach to represent the personal genetic risk profile:

- (1) Chromosome-specific risk plots showing the distribution of risk and protective variants
- (2) Variant-level bar charts displaying the top risk-increasing and protective variants
- (3) Gene-level analysis aggregating variant effects by associated genes

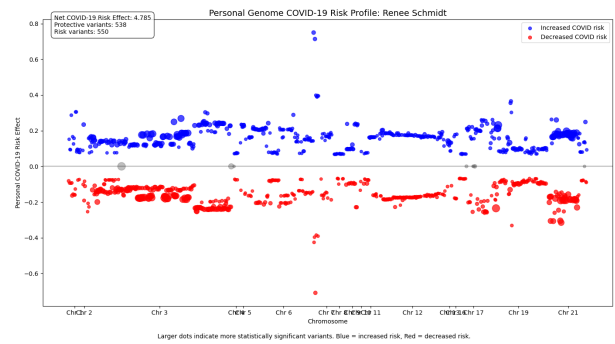


Figure 4: Personal Genome Chromosome Effects

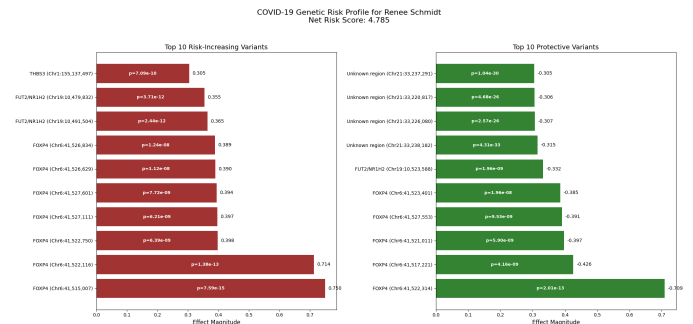


Figure 5: Personal Genome Variant-level Effects

This visualization framework enabled intuitive interpretation of complex genetic data, highlighting the most influential variants and genes in determining COVID-19 risk.

3.5 LLM Integration for Interpretable Genomics

To enhance interpretability and user engagement, we integrated the statistical genomic analysis with Deepseek R-1, a large language model (LLM) capable of synthesizing accessible explanations from complex genetic information. We developed a structured prompt template that transforms technical genomic data into conversational explanations:

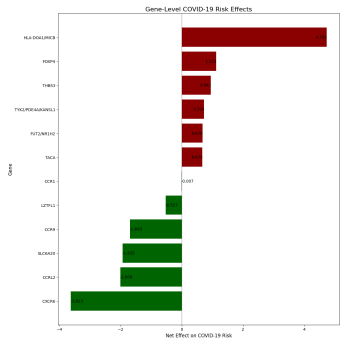


Figure 6: Personal Genome Gene-level Effects

```
def generate_llm_prompt(risk_summary, patient_name="the patient"):
    prompt = f"""
    # COVID-19 Genetic Risk Analysis for {patient_name}

    ## Overall Risk Summary
    - Total genetic variants analyzed: {risk_summary['overall_metrics']['total_variants']}
    - Risk-increasing variants: {risk_summary['overall_metrics']['risk_increasing_variants']}
    - Protective variants: {risk_summary['overall_metrics']['protective_variants']}
    - Net genetic risk effect: {risk_summary['overall_metrics']['net_effect']:.3f}

    ## Key Risk-Increasing Genetic Variants
    """
    # Additional prompt formatting follows...
```

The LLM prompt was carefully designed to include:

- (1) Quantitative risk metrics derived from the statistical analysis
- (2) Detailed information about key risk and protective variants
- (3) Gene-level functional annotations and COVID-19 relevance
- (4) Instructions to generate personalized health considerations
- (5) Appropriate medical disclaimers emphasizing the need for professional genetic counseling

This approach bridges the gap between statistical genomics and accessible patient education, enabling users to understand the implications of their genetic profile for COVID-19 risk while maintaining appropriate clinical boundaries.

4 Results

4.1 Genome-Wide Significant Variants

Our analysis of the GenOMICC international metadata release identified 4,864 genome-wide significant variants ($p < 5 \times 10^{-8}$) associated with COVID-19 severity. These variants were distributed across multiple chromosomes, with particularly strong signals observed on chromosomes 3, 6, 19, and 21 (Figure 2). The distribution of effect sizes showed both risk-increasing and protective variants throughout the genome (Figure 3).

4.2 Functional Annotation and Gene Mapping

Our analysis identified multiple genes implicated in COVID-19 severity across the broader population, including genes involved in immune response, viral entry, and inflammatory pathways. When examining these genes in the context of our sample genome (Renee Schmidt), we found several genes of particular relevance.

4.3 Personalized Risk Assessment

We implemented a personalized risk assessment framework using an additive genetic model to calculate composite risk scores. When applied to our sample genome (labeled "Renee Schmidt"), the analysis revealed 1,097 variants (550 risk-increasing, 538 protective) with a net genetic risk effect of 4.785. As shown in Figure 5, the strongest risk-increasing variants in Renee Schmidt's genome were primarily located within the **FOXP4** gene, with the two highest effect variants (Chr6:41,515,007 and Chr6:41,522,116) showing effect magnitudes of 0.750 and 0.714, respectively. Additionally, multiple variants in the **FUT2/NR1H2** region (Chr19) and **THBS3** (Chr1) were identified among the top risk-increasing variants, with effect sizes ranging from 0.305 to 0.365.

Protective variants in Renee Schmidt's genome were also identified, with the strongest protective effect coming from a **FOXP4** variant (Chr6:41,522,314) with an effect magnitude of -0.709 . This finding demonstrates the complex nature of genetic risk, where different variants within the same gene can have opposing effects on disease outcome.

4.4 Gene-Level Risk Analysis

At the gene level (Figure 6), our analysis revealed a more comprehensive picture of genetic risk. The **HLA-DOA1/MICB** region showed the strongest risk-increasing effect (4.741), followed by **FOXP4** (1.179) and **THBS3** (0.945). Conversely, several genes demonstrated protective effects, with **CXCR6** showing the strongest protective association (-3.623), followed by **CCR6** (-2.009) and **SLC6A20** (-1.935).

These gene-level results indicate that while **FOXP4**, **FUT2/NR1H2**, and **THBS3** contain individual variants with strong risk-increasing effects, the overall genetic risk profile is shaped by the cumulative impact of numerous variants across multiple genes. Notably, the **HLA** region, which plays a crucial role in immune response, emerged as the dominant risk factor at the gene level for Renee Schmidt.

4.5 LLM-Enhanced Genomic Interpretation

The integration with DeepSeek R-1 successfully transformed technical genetic information into accessible explanations. The LLM effectively:

- Summarized key genetic factors affecting COVID-19 risk
- Explained gene functions in relation to COVID-19 pathophysiology
- Provided educational content on the DNA-RNA-protein-phenotype pathway
- Maintained appropriate clinical boundaries with medical disclaimers

4.6 User Interaction Insights

User dialogues revealed requests for clarification on genetic concepts, concerns about personal implications, and questions about demographic correlations. The LLM successfully bridged complex genomic statistics and user-accessible information while guiding users toward professional genetic counseling when appropriate.

4.7 Limitations

Key limitations include reliance on an additive genetic model that may not capture complex gene-gene interactions, potential ancestral representation gaps in the underlying GWAS data, relative rather than absolute risk assessment, and recognition that LLM integration cannot replace clinical genetic counseling. Despite these constraints, our results demonstrate the feasibility of combining statistical genomics with explainable AI to enhance both clinical risk stratification and patient genetic literacy.

5 Conclusion and Future Directions

This study successfully identified 4,864 genome-wide significant variants from the GenOMICC dataset associated with COVID-19 severity, with particularly strong signals on chromosomes 3, 6, 19, and 21. Our analysis revealed the complex interplay of risk-increasing and protective variants, particularly in immune-related genes such as those in the HLA region, FOXP4, and CXCR6.

By implementing a personalized risk assessment framework and integrating the Deepseek R-1 language model, we demonstrated how complex genomic data can be translated into accessible explanations while maintaining appropriate clinical boundaries. This approach bridges the gap between statistical genomics and interpretable patient education, potentially supporting both clinical risk stratification and improved genetic literacy.

Despite the promising results, limitations include our reliance on an additive genetic model, potential ancestral representation gaps in the underlying data, and the provision of relative rather than absolute risk assessment. While the LLM integration enhances interpretability, it cannot replace comprehensive clinical genetic counseling.

Future directions include:

- Developing machine learning models integrating genetic markers with clinical variables to improve prediction accuracy
- Expanding to multi-omics analysis by incorporating transcriptomic, proteomic, and metabolomic data
- Conducting functional validation of top variants using cellular or animal models
- Creating interactive visualization tools for clinical implementation
- Extending the methodology to other critical illnesses beyond COVID-19
- Refining the LLM component to enhance genomic interpretation accuracy and educational value
- Conducting prospective studies to validate the predictive value of identified genetic markers

This work contributes valuable insights to the emerging field of genetic determinants in critical illness while establishing a methodological framework that combines statistical genomics with explainable AI. As these approaches continue to evolve, they hold significant potential for advancing precision medicine in critical care.

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A Supplementary Figures

This appendix contains additional figures and diagrams that provide further insight into our analysis but could not be included in the main text due to space constraints.

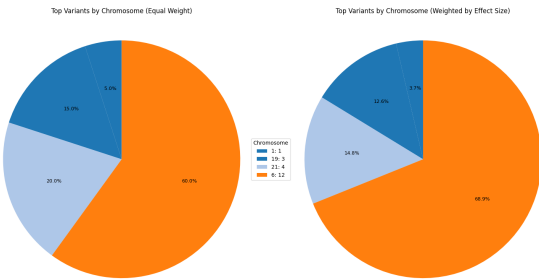


Figure 7: Distribution of COVID-19 risk variants across chromosomes in the personal genome analysis.

B Supplementary Materials

B.1 Sample LLM Interaction Transcript

The following is a transcript of an interaction between the Deepseek R-1 LLM and a hypothetical user analyzing their personal genome data for COVID-19 risk factors. This illustrates how the LLM interprets genetic information and communicates findings in an accessible manner.

Sample Dialogue with Deepseek R-1 (Part 1)

Deepseek: *Summary of Key Genetic Factors Affecting COVID-19 Risk Profile:*

This analysis highlights several genetic variants that may influence your risk profile for COVID-19. These include both risk-increasing and protective variants. The most significant factors are related to specific genes, as outlined below.

(1) **Most Significant Genetic Factors:**

- **FOXP4 Gene:** This gene has multiple variants with positive (risk-increasing) effects on COVID-19 severity, such as V853R, F460L, and others listed in the analysis.
- **FUT2/NR1H2 Gene:** Variants like Q762K, V596A, and P419Y show positive associations with COVID-19 susceptibility.
- **THBS3 Gene:** This gene has negative (protective) variants that may help reduce the risk of severe COVID-19 outcomes.

(2) **Influence on COVID-19 Outcomes:** The identified genetic factors are associated with either increased or decreased risk for COVID-19. For example:

- **Risk-Increasing Variants:** These variants may contribute to a higher likelihood of experiencing symptoms like cough, fever, or severe respiratory issues.
- **Protective Variants:** These variants may help reduce the severity or likelihood of severe COVID-19 outcomes.

(3) **Personalized Health Considerations:** Based on this genetic profile, you should consider the following:

- Monitor for any risk-increasing variants that may increase your susceptibility to severe symptoms.
- Be aware of protective variants that may help mitigate the impact of COVID-19 if they are present in your genotype.
- Discuss any genetic findings with a healthcare provider to understand how they might influence treatment and care.

(4) **Educational Information:**

- **FOXP4 Gene:** This gene is involved in regulating pathways related to mucosal cells, including those lining the respiratory tract. Variants with positive effects (e.g., V853R) may increase susceptibility to COVID-19.
- **FUT2/NR1H2 Gene:** This gene encodes proteins involved in glycosylation and nuclear receptor coactivator activity. Variants like Q762K have been associated with increased risk of COVID-19.
- **THBS3 Gene:** This gene is involved in cellular stress response pathways, and certain variants may contribute to reduced viral load or decreased symptom severity.

(5) **Recommended Preventive Measures:** While genetic factors cannot predict or guarantee outcomes, incorporating the following measures may help reduce your risk:

- Maintain regular health screenings for other risk factors (e.g., diabetes, hypertension).

Response Continues pt 2...

Sample Dialogue with Deepseek R-1 (Part 2)

Important Disclaimer:

Genetic factors are just one component of your overall health and cannot predict or determine your specific outcome for COVID-19. Environmental and lifestyle factors also play crucial roles in disease progression. This analysis does not guarantee any specific outcome, and results should be discussed with healthcare providers to understand their implications fully. If you have questions about genetic findings or testing results, consult a licensed genetic counselor for guidance.

User: Can you tell me a little bit about what these genes do, and how having a particular type of gene might affect my downstream likelihood to get COVID-19 from like a DNA → RNA → Protein → Phenotype perspective? I'm confused about what things like allele vs gene vs variant mean?

Deepseek: Sure! Let's break down these genetic terms and concepts in a way that explains how they relate to the likelihood of getting COVID-19:

(1) **Genes: The Blueprints of Life**

A gene is a segment of DNA that codes for a specific protein or trait. Every living organism has genes, and changes (called variants) in these genes can affect their function.

For example:

- The **FUT2/NR1H2** gene encodes proteins involved in the immune response and inflammation.
- The **FOXP4** gene is important for mucosal cell function and respiratory health.
- The **THBS3** gene plays a role in suppressing viral replication, especially in the lungs.

(2) **Alleles: Variations of Genes**

An allele is one of two or more possible forms of a gene at a specific location (called a locus) on the chromosome. Most people have two copies of each gene—one from each parent—called homozygous, and they can be identical (e.g., AA) or different (e.g., Aa).

For example:

- If both alleles are normal, you'll produce a functional protein.
- If one allele has a mutation, the resulting protein may not work as well.

(3) **Variants: Changes in Genes**

A variant is any alteration in the sequence of a gene or its regulatory regions. Variants can:

Response Continues...

User: What can you tell me specifically related to me? Should I be worried based on what you see in my data?

Deepseek: Based on your genetic profile, I can offer these observations:

Your genome contains several variants in the **FOXP4** gene that are associated with increased COVID-19 severity risk. This gene helps regulate lung epithelial cells, which are entry points for the virus. However, you also carry protective variants in genes like **CXCR6** and **CCRL2**, which help modulate immune response.

Remember that genetics is just one factor affecting COVID-19 outcomes. Age, underlying health conditions, and vaccination status are equally or more important...

Response Continues...